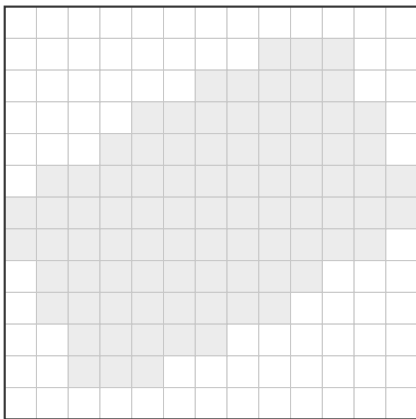


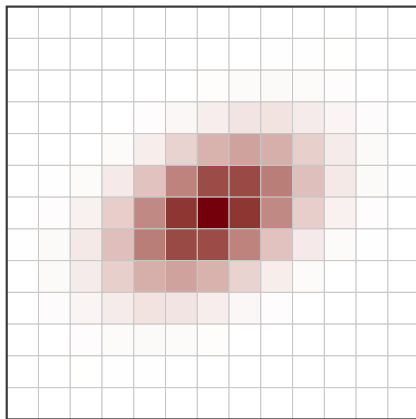
Rectangular Kernel Construction

(A) Mask M_R



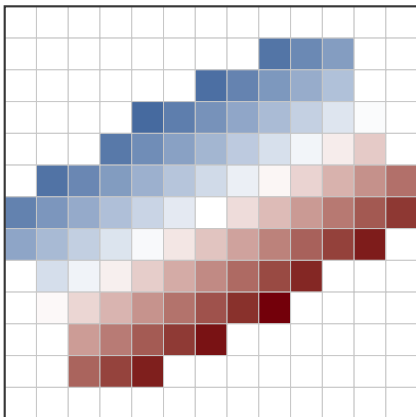
$$M_R(i, j) = \mathbb{1}[|u| \leq 3\sigma_u, |v| \leq 3\sigma_v]$$

(B) Envelope $G(u, v)$

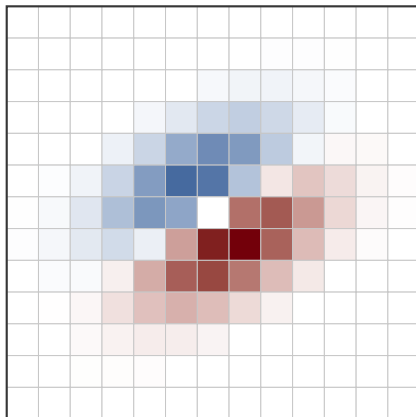


$$G(u, v) = \exp\left(-\frac{1}{2}\left(\frac{u^2}{\sigma_u^2} + \frac{v^2}{\sigma_v^2}\right)\right)$$

(C) Derivative $-v$



(D) Combined K_R



$-v$ profile: positive (garnet) / negative (blue)

$$K_R = -v \cdot G(u, v) \cdot M_R$$